Introduction to Variational Autoencoders and their applications.

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Overview

1. Introduction
2. Variational Auto-Encoder - The Idea
   - Auto-Encoder
3. VAE framework
   - The idea
   - Mathematical foundations
   - Changing the model
   - Tweaking the approximate posterior distribution
4. Toward a Geometry-Aware VAE
   - The framework
   - The proposed model
   - A new way to generate data
   - Sensitivities and robustness on toy data
5. Results on Neuroimaging data
   - Materials
   - Methods
   - Results

Initial goal: Dimension reduction

Classical methdods for dimensionality redictions (PCA, ICA, etc..) are

- linear $\rightarrow$ may not be able to capture correct features,
- constrained by simple probabilistic models $\rightarrow$ may not describe the population well.

Need for non linear methods! Auto-encoders emerged as one possible solution...

With many other applications!

Auto-Encoder

- The objective $\Longrightarrow$ Dimensionnality Reduction

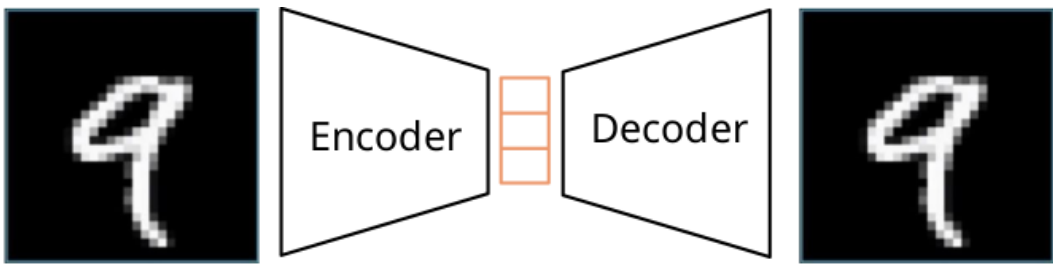


Figure: Simple Auto-Encoder

AutoEncoder

Assumptions:

- Let $x \in \mathcal{X}$ be a set a data. We assume that there exists $z \in \mathcal{Z}$ such that z is a low dimensional representation of x
- The encoder e_θ and decoder d_ϕ are functions modelled by neural networks (NNs) such that θ and ϕ are the weights of the NNs
- Let x' be the reconstructed samples, the objective is to have $x \simeq x'$

The Objective function writes:

$$\mathcal{L} = \|x - x'\|^2 = \|x - d_\phi(z)\|^2 = \|x - d_\phi(e_\theta(x))\|^2$$

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$\Longrightarrow$ The networks are optimised using stochastic gradient descent

$$\phi \leftarrow \phi - \varepsilon \cdot \nabla_\phi \mathcal{L}$$
$$\theta \leftarrow \theta - \varepsilon \cdot \nabla_\theta \mathcal{L}$$

AutoEncoder - Shortcomings

- What is represented in the latent space?

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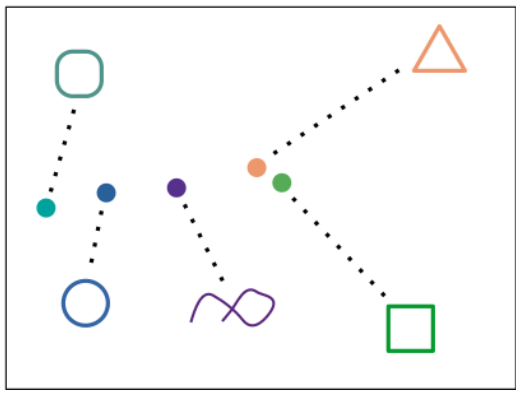


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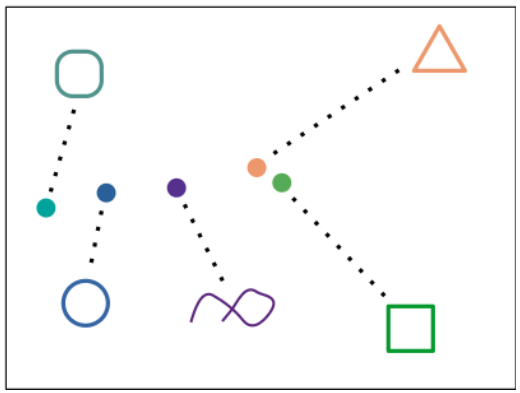


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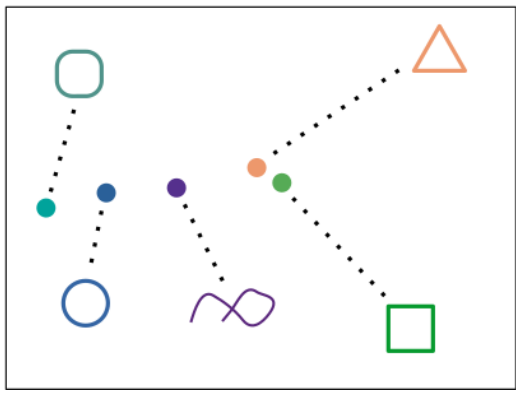


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$\Longrightarrow$ Need for a new framework

VAE - The Idea

- An auto-encoder based model...

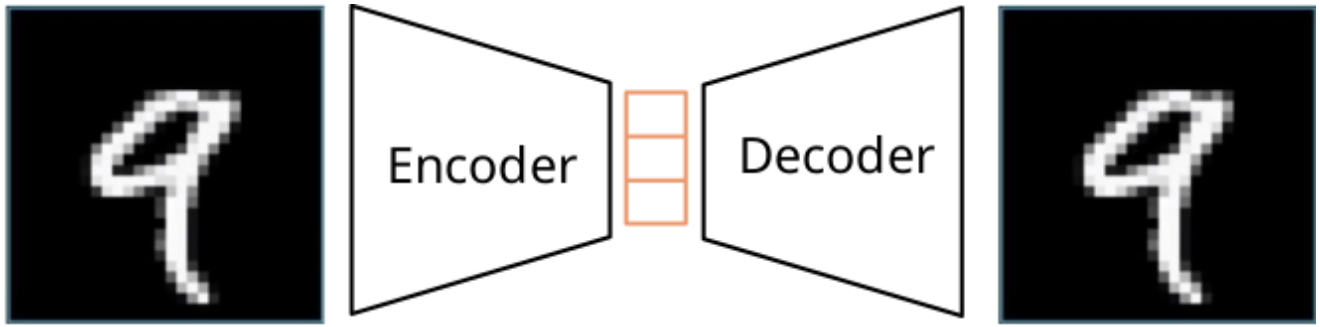


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- ... but where an input data point is encoded as a **distribution** defined over the latent space [KW14, RMW14]

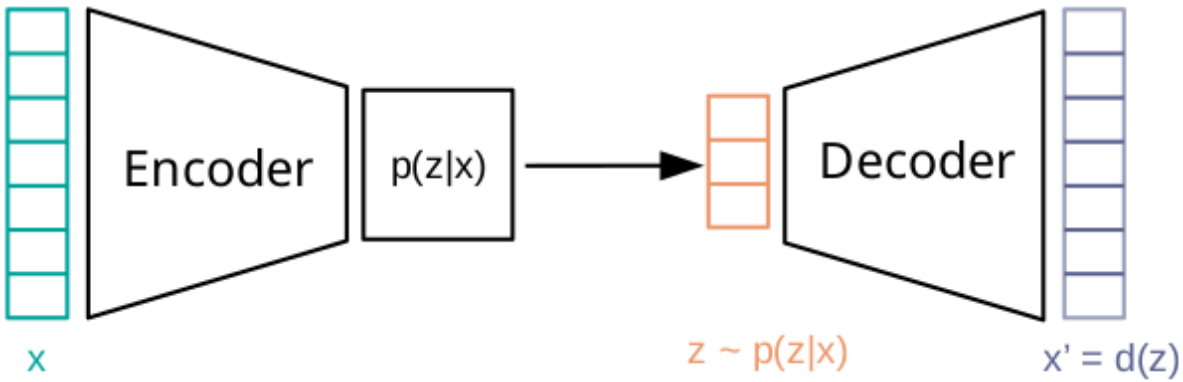


Figure: VAE framework

VAE - Mathematical Considerations

- Let $x \in \mathcal{X}$ be a set of data and $\{P_\theta, \theta \in \Theta\}$ be a parametric model
- We assume there exists latent variables $z \in \mathcal{Z}$ living in a smaller space such that the marginal likelihood writes

$$p_\theta(x) = \int p_\theta(x|z) q_{\text{prior}}(z) dz \,,$$

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Problem: The integral is often intractable.

Variational inference

We have to use Variational Inference:

$$\log p_\theta(x) \quad = \quad \log \left(\int p_\theta(x|z) q_{\text{prior}}(z) dz \right)$$

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&\geq \int (\log p_\theta(x,z))\, q(z) dz - H(q(z))
\end{aligned}
$$

with H the entropy of $q(z)$.

The equality holds for $q(z) = q_\theta(z|x)$.

Variational inference: The ELBO

- Well-know issue: the posterior $q(z) = q_\theta(z|x)$ is intractable.

  $\longrightarrow$ use Expectation-Maximization like algorithms (MCMC-SAEM version if needed)
- **OR** approximate this posterior:

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  This leads to an unbiased estimate of the log-likelihood

$$\widehat{p_\theta}(x) = \frac{p_\theta(x,z)}{q_\phi(z|x)}, \qquad \mathbb{E}_{z\sim q_\phi(z|x)}[\widehat{p_\theta}(x)] = p_\theta(x)\,,$$

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and the definition of the **Evidence Lower Bound** (ELBO):

$$\begin{aligned}\log p_\theta(x) &\geq \mathbb{E}_{z\sim q_\phi(z|x)}[\log(p_\theta(x,z)) - \log(q_\phi(z|x))] \\ &\geq \mathit{ELBO}\end{aligned}$$

Variational inference: The ELBO

Objectives:

1. Optimize the ELBO **as a function** instead of the target distribution

   Use stochastic gradient descent in both θ and ϕ

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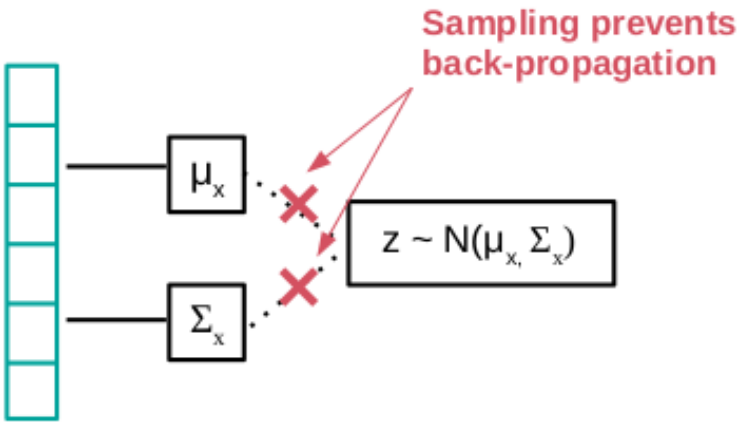
   Use stochastic gradient descent in both θ and ϕ

2. Optimize the ELBO **as a bound** to get closer to the target

   Use sampling methods to produce samples $z \sim q_\theta(z|x)$

The Reparametrization Trick for stochastic gradient descent

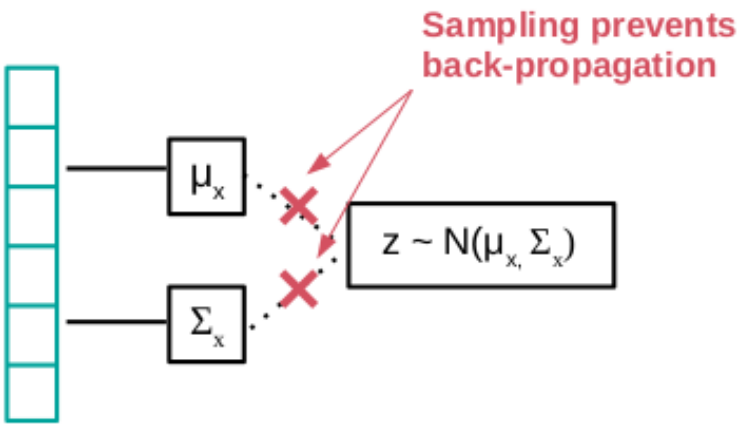
- Since $z \sim \mathcal{N}(\mu_\phi(x), \Sigma_\phi(x))$, the model is not amenable to gradient descent



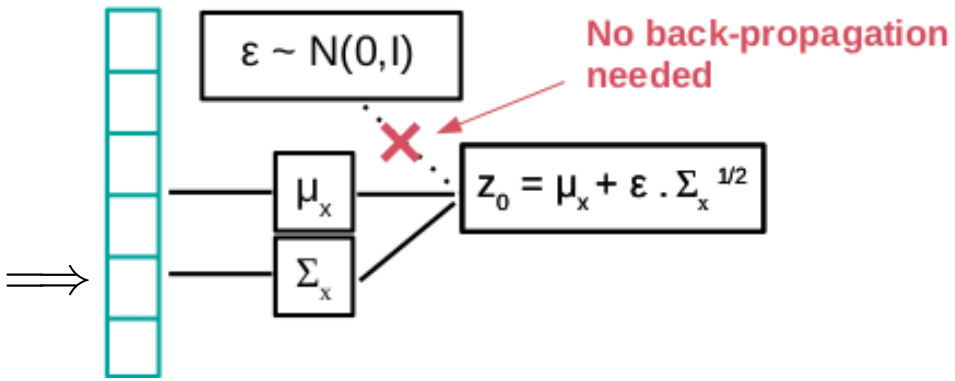
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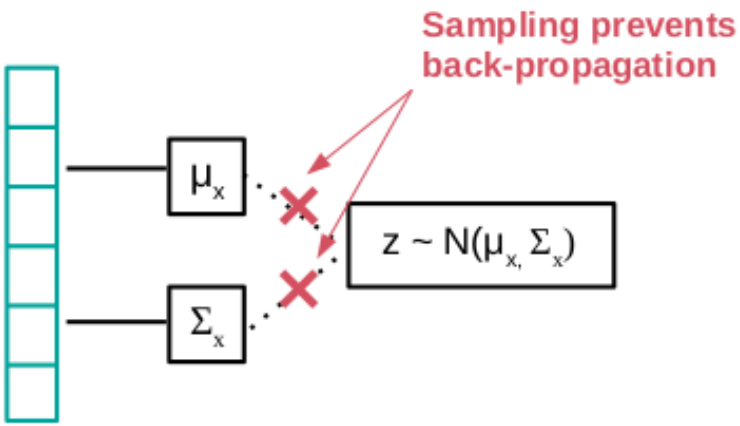
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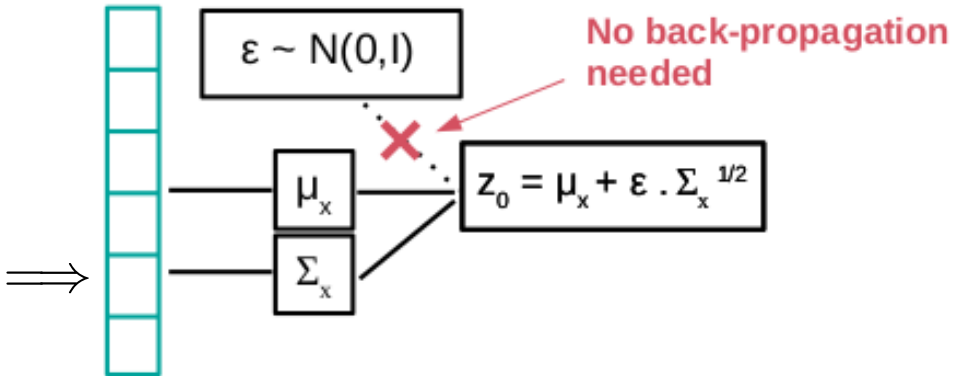
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$\Longrightarrow$ Optimization with respect to encoder and decoder parameters made possible !

Objective 1.



Changing the model

General VAE model requires the choice of $q_{prior}(z)$.

$\rightarrow$ Most common choice: $q_{prior}(z) = \mathcal{N}(0, 1)$.

Pros:

- good intial gess,
- enables easy computations

Cons:

- very restrictive as it is isotropic and monomodal

Can we improve the model itself choosing other distributions?

Other VAE models changing the prior

Literature is full of propositions:

- the first one being a mixture of Gaussian distribution ("Approximate inference for deep latent gaussian mixtures", "Deep Unsupervised Clustering with Gaussian Mixture Variational Autoencoders")
- VAE with a VampPrior
- up to prior leaning (VQVAE, *RAE_L2*)

Requires to be able optimize the ELBO: many limitations!

Other VAE models changing the distance in the ELBO

One may prefer to change the definition of the distance between probability distributions.

- Weighting the KL−divergence in the ELBO ($\beta - VAE$)
- Changing from KL−divergence to Wassenstein distance between the data distribution and the model one (WAE).

Other VAE models changing the Reconstruction metric

One may not want to use the standard L^2 norm on images.

- metric by patch and no more pixelwise (MSSSIM_VAE)
- Discriminator extract features and the similarity is between extracted features (VAEGAN)

Tweaking the Approximate Posterior Distribution

Concerning Objective 2.

- The ELBO can written as

$$ELBO = \log p_\theta(x) - \underbrace{\mathrm{KL}(q_\phi(z|x)||p_\theta(z|x))}_{\approx 0 \text{ if } q_\phi(z|x) \approx p_\theta(z|x)} .$$

- Kullback-Leiber divergence $\geq 0 \Rightarrow$ make it vanish by tweaking the approximate posterior $q_\phi(z|x)$

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- How? and how to ensure that the model would still be amenable to the back-propagation ?

Solution 1: Normalizing Flows

- Use smooth invertible parametrized mappings f_ψ to "sample" z [RM15]

- Apply K transformations to $z_0 \sim q_{init}$ (here $q_{init} = q_\phi$)
- Final random variable $z_K = f_x^K \circ \cdots \circ f_x^1(z_0) \sim q_\phi(z_K|x)$ with

$$q_\phi(z_K|x) = q_\phi(z_0|x) \prod_{k=1}^{K} |\det \mathbf{J}_{f_x^k}|^{-1}, \tag{1}$$

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Objective 2.



Many references (VAE_LinNF, VAE_IAF)

although difficult to compute the Jacobian of these maps f_1^K

Solution 2: Hamiltonian VAE

- Idea = Hybrid Monte Carlo Sampler [No11, DMS17, LBB+19],
- Target density

$$p_\theta(z|x) = \frac{p_\theta(x,z)}{p_\theta(x)} \propto p_\theta(x,z) = \pi_x(z)\,.$$

- Introduce an auxiliary random variable $\rho \sim \mathcal{N}(0, \mathbf{M})$ called "momentum"
- Write the Hamiltonian:

$$\begin{aligned} H_x(z,\rho) &= -\log \pi_x(z,\rho) \\ &= -\log \pi_x(z) + \frac{1}{2}\log((2\pi)^d|\mathbf{M}|) + \rho^\top \mathbf{M}^{-1}\rho \\ &= U_x(z) + \kappa(\rho)\,. \end{aligned}$$

- Sample (z,ρ) with this dynamic.

Solution 2: Hamiltonian VAE

- Use a discretization scheme

$$\begin{aligned}
\rho(t+\varepsilon/2) &= \rho(t) - \frac{\varepsilon}{2} \cdot \nabla_z H(z(t), \rho(t)) \,, \\
z(t+\varepsilon) &= z(t) + \varepsilon \cdot \nabla_\rho (H(z(t), \rho(t+\varepsilon/2))) \,, \\
\rho(t+\varepsilon) &= \rho(t+\varepsilon/2) - \frac{\varepsilon}{2} \cdot \nabla_z H(z(t+\varepsilon), \rho(t+\varepsilon/2)) \,,
\end{aligned} \tag{2}$$

- A proposal $(\widetilde{z}, \widetilde{\rho})$ is accepted with probability:

$$\alpha = \min\Big(1, \exp\big(-H(\widetilde{z}, \widetilde{\rho}) + H(z, \rho)\big)\Big)$$

$\Longrightarrow$ Creates an ergodic, time-reversible Markov Chain having π_x as stationary distribution.
Note that the Metropolis Hastings' acceptation step has to be removed for the back propagation to be possible.

Hamiltonian VAE

- The graphical scheme [CDS18]

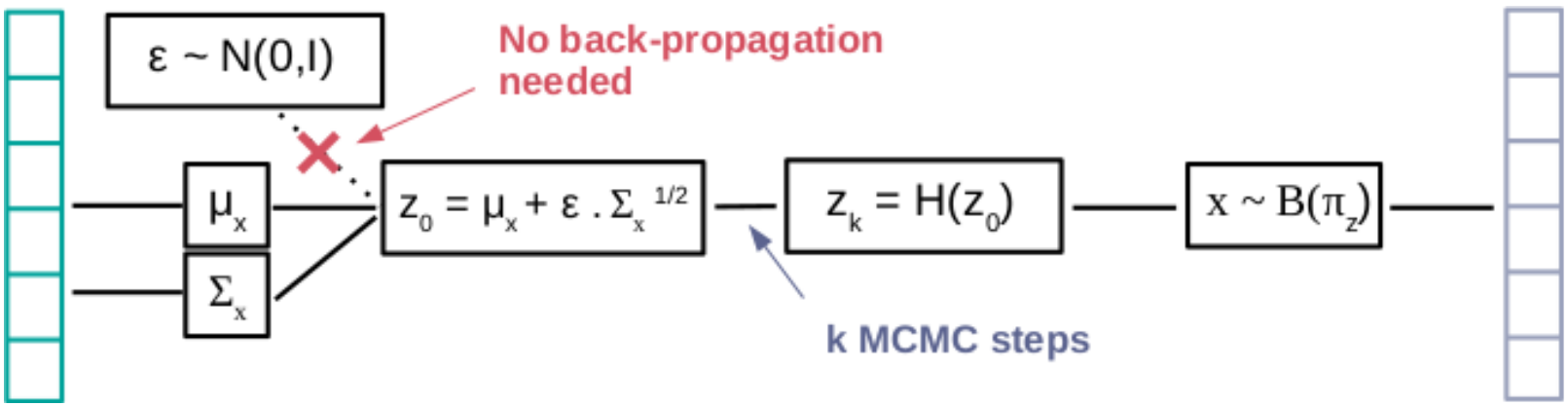


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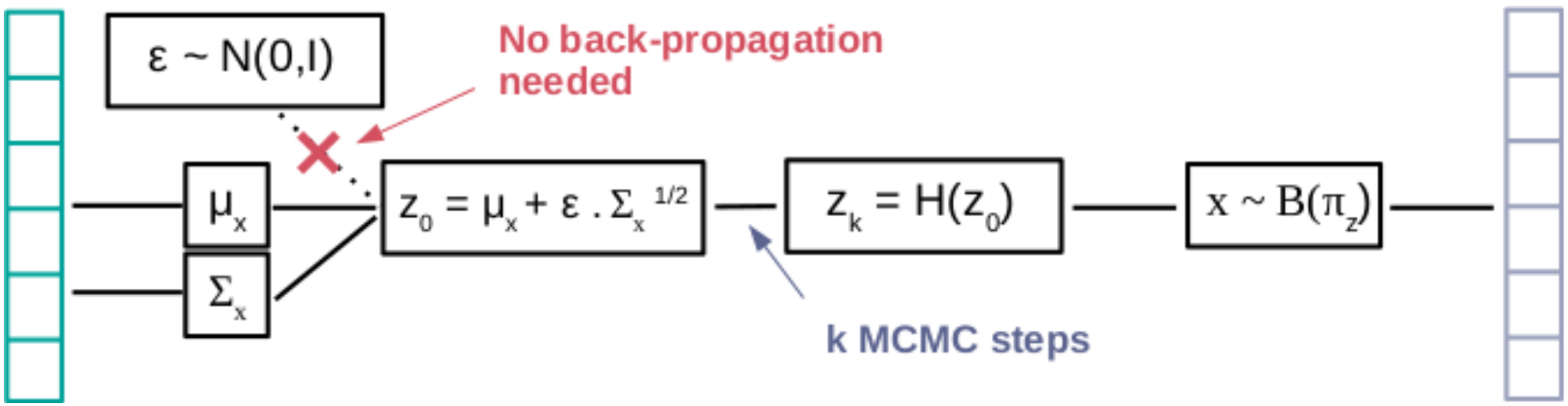


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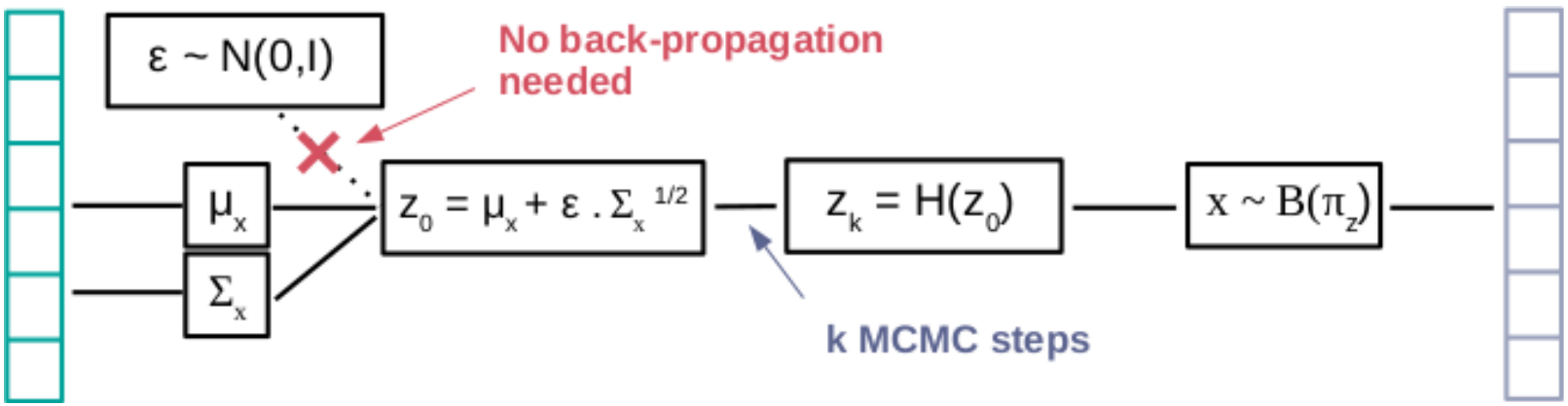


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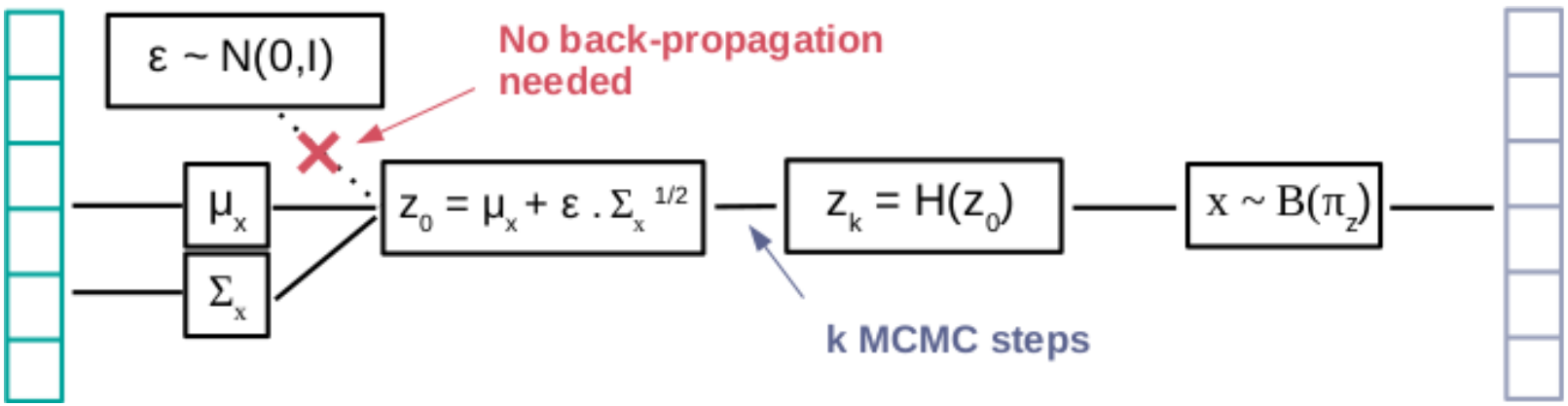


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What about geometry?

Benchmark of many of these methods

For an extensive comparison of the VAE and VAE like models:

https://github.com/clementchadebec/benchmark_VAE

- Benchmark of the learning strategies
- Benchmark of the sampling strategies

Defining a New Framework

Assumptions:

- As of now the latent space structure was supposed to be Euclidean (i.e. $\mathcal{Z} = \mathbb{R}^d$)
- Let us now relax this hypothesis and assume that $\mathcal{Z}$ is a Riemannian manifold endowed with a metric $\mathbf{G}$.
- It was shown that exploiting the geometrical aspect of probability distributions can lead to far more efficient sampling [GCC09, GC11]

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Our ideas:

1. Exploit the manifold structure of the latent space to improve the posterior sampling
2. Learn the metric defined in the latent space
3. Use the learned geometry to generate instead of the prior [CTSBA21]

Riemanian geometry principles

- Riemanian manifold: (reduced to our model) $\mathbb{R}^d$ endowed with a metric $\mathbf{G}$: $\mathcal{M} = (\mathbb{R}^d, \mathbf{G})$.
  $\Longrightarrow$ $\mathbb{R}^d$ not flat anymore, curved space (as montains)

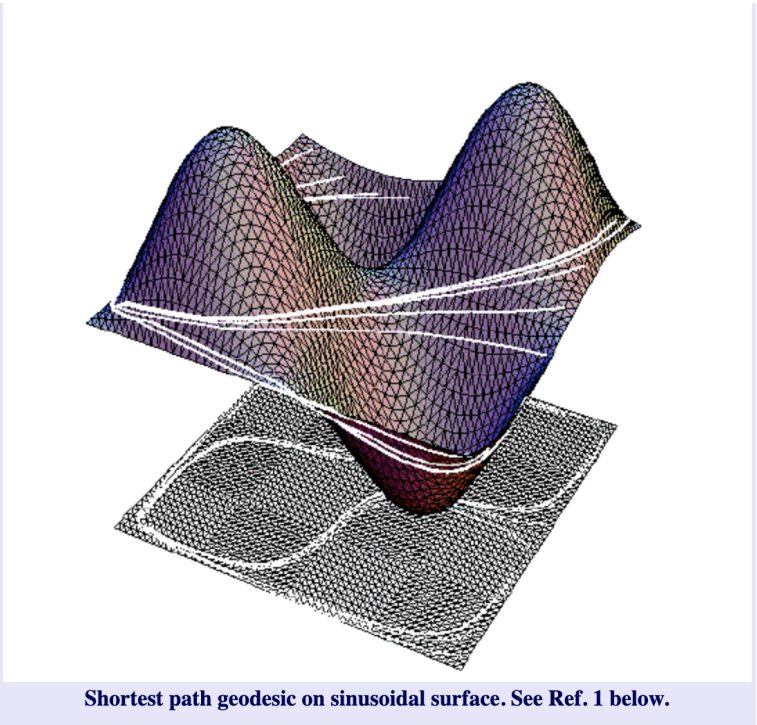


Figure: Image taken from: Fast Marching Methods on Triangulated Domains : Kimmel, R., a Sethian, J.A., Proceedings of the National Academy of Sciences, 95, pp. 8341-8435, 1998

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- Geodesic curves:
  - Length of a curve $\gamma : [0, 1] \to \mathcal{M}$ from z_1 to z_2 living in a Riemannian manifold $\mathcal{M}$

$$L(\gamma) = \int_0^1 \sqrt{\langle \gamma'(t), \gamma'(t) \rangle_{\gamma(t)}} dt \qquad \gamma(0) = z_1, \gamma(1) = z_2 \,. \tag{3}$$

  - Geodesic paths = curve γ minimizing Eq. (3)
  - or equivalently minimizing the curve energy

$$E(\gamma) = \int_0^1 \langle \gamma'(t), \gamma'(t) \rangle_{\gamma(t)} dt \qquad \gamma(0) = z_1, \gamma(1) = z_2 \,.$$

1) Improve Posterior Sampling - Riemannian HMC

- Rely on the Riemannian Hamiltonian Monte Carlo Sampler [GC11]:
  - Introduce a Position-specific random momentum $\rho \sim \mathcal{N}(0, \mathbf{G}(z))$
  - Simulates the evolution $(z(t), \rho(t))$ of a particle whose motion is governed by Hamiltonian dynamics on the manifold
- The Hamiltonian writes

$$H_x^{Riem}(z, \rho) = \log p_{\text{target}}(z) + \frac{1}{2} \log((2\pi)^D \det \mathbf{G}(z)) + \frac{1}{2} \rho^\top \mathbf{G}(z)^{-1} \rho \,.$$

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- Use of the "Generalized" Leapfrog integrator to sample from p_{target}

Pros:

- Use the underlying geometry of the data to improve sampling

Cons:

- The metric is unknown

2) Learn the Metric - The Choice of the Metric

- Parametric metric: [Lou19]:

$$\mathbf{G}^{-1}(z) = \sum_{i=1}^{N} L_{\psi_i} L_{\psi_i}^{\top} \exp\left(- \frac{\|z - c_i\|_2^2}{T^2} \right) + \lambda I_d \,,$$

- L_{ψ_i} lower triangular matrices parametrized using neural networks
- T temperature to smooth the metric
- c_i centroids
- λ regularization factor

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Pros:

- Closed-form expression of the inverse metric $\Longrightarrow$ useful for geodesic computation
- Geodesics travel through most populated areas.

The Model - Riemannian Hamiltonian VAE

- The graphical scheme

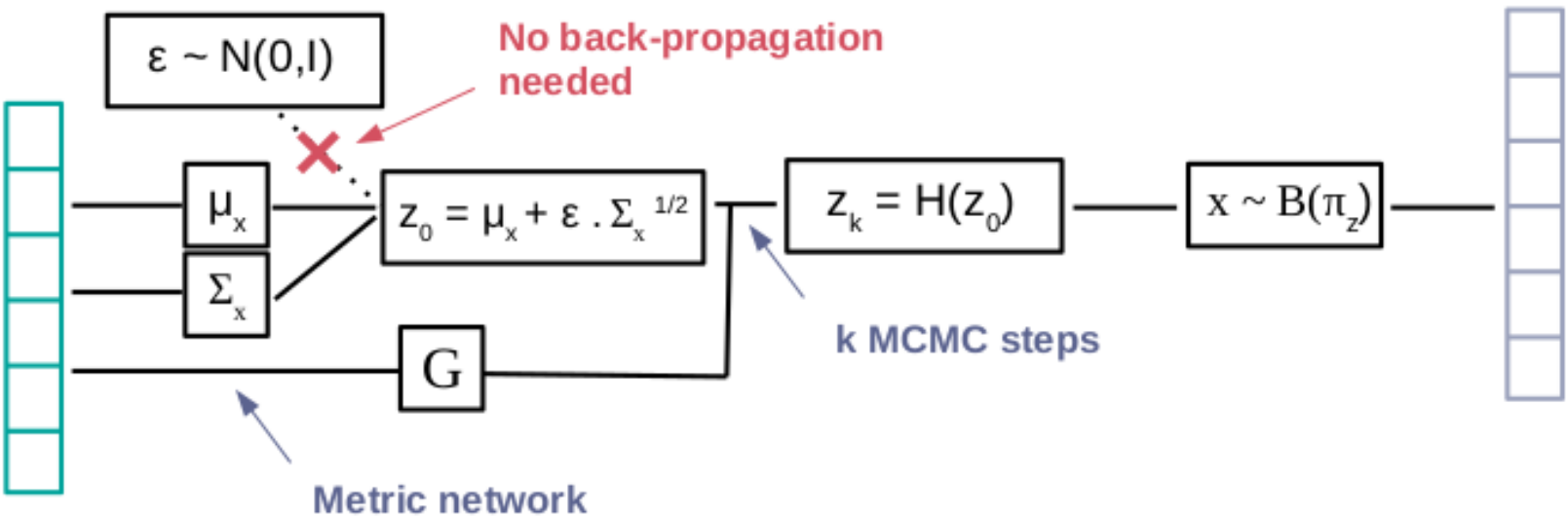
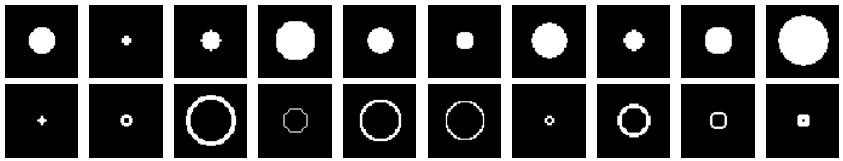


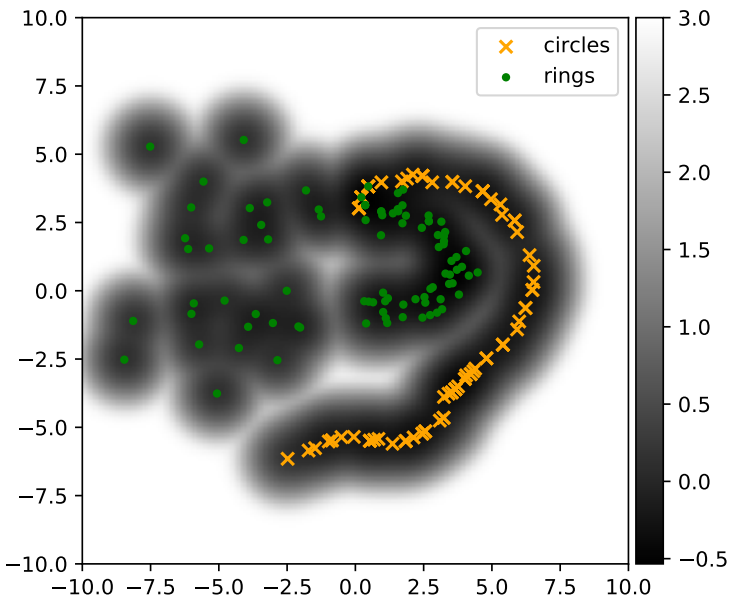
Figure: Riemannian Hamiltonian VAE.

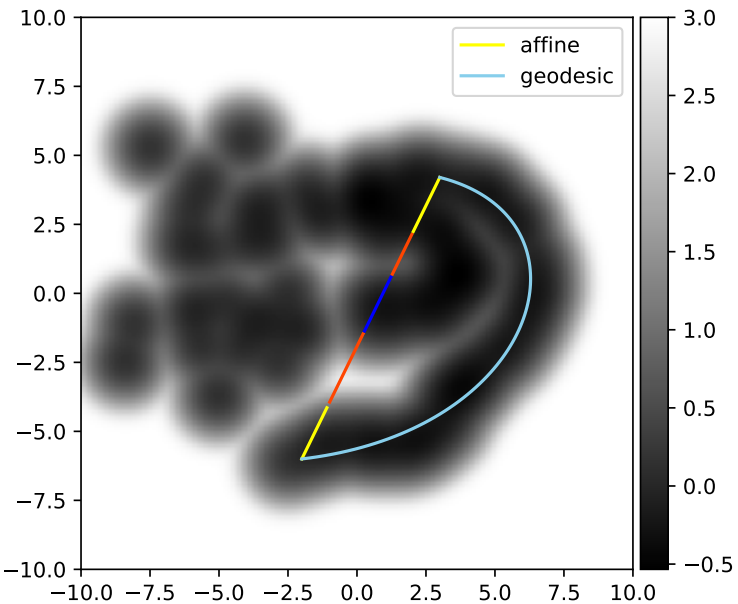
The Learned Latent Space examples

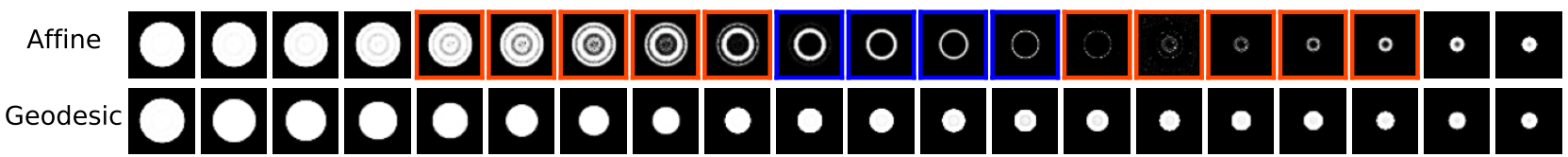
Training samples:



Latent space and interpolations:

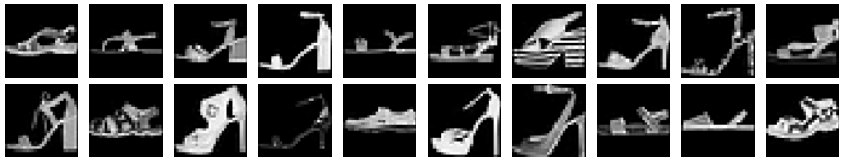




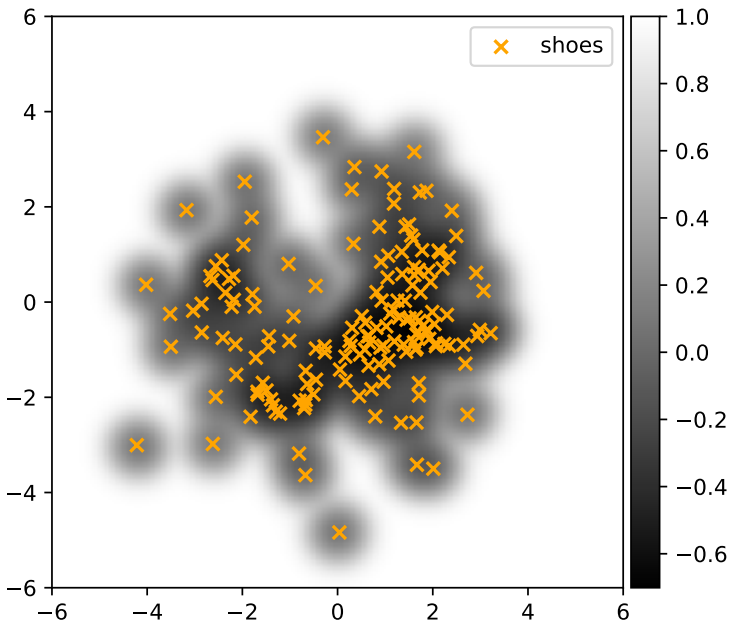


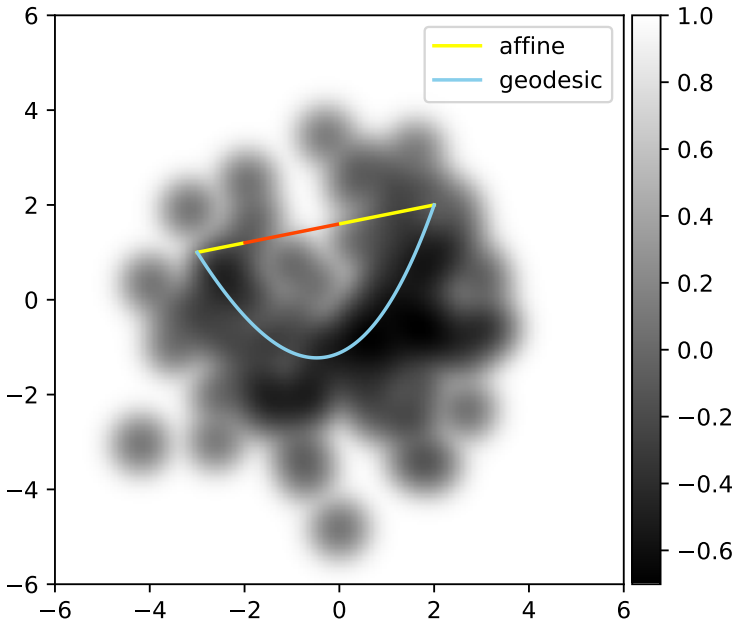
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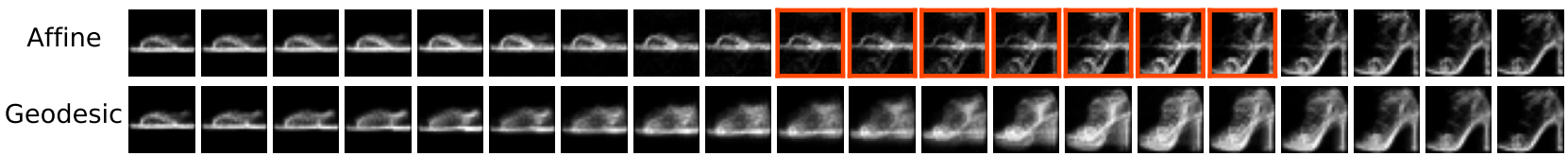
Training samples:



Latent space and interpolations:







3) Improve Data Generation - Sample With the Metric

Idea:

- Use a geometry-based sampling procedure: pdf driven by the metric

$$p(z) = \frac{\mathbb{1}_S(z)\sqrt{\det \mathbf{G}^{-1}(z)}}{\int_{\mathbb{R}^d} \mathbb{1}_S(z)\sqrt{\det \mathbf{G}^{-1}(z)dz}},$$

  where S is a compact set and $\mathbb{1}_S(z) = 1$ if $z \in S$, 0 otherwise.
- Use of classic MCMC sampler (e.g. Hamiltonian Monte Carlo)

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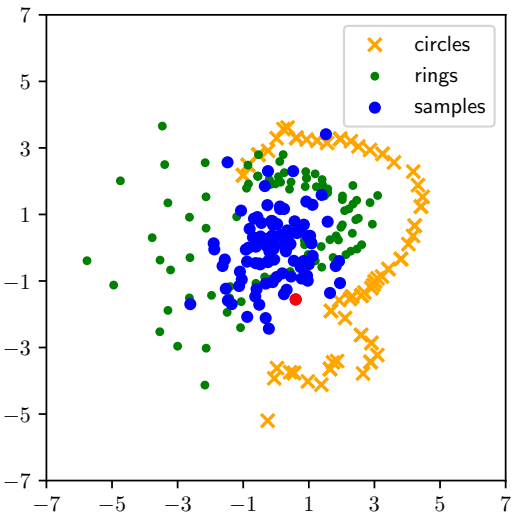
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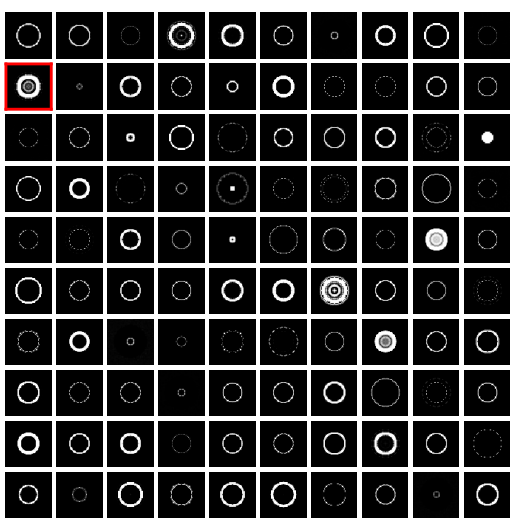
Pros:

- $\mathbf{G}^{-1}$ easily computable
- Samples "close" to the data

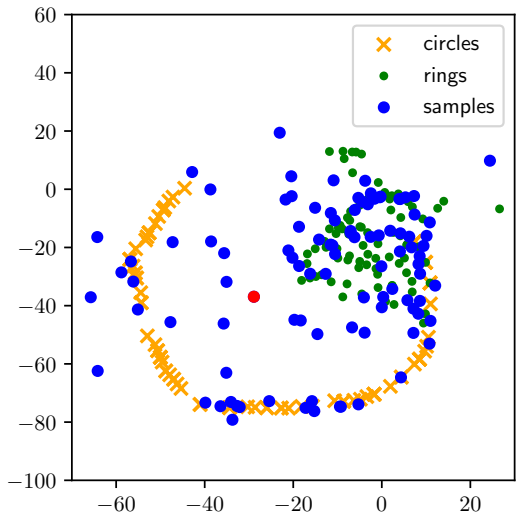
Sampling Comparison

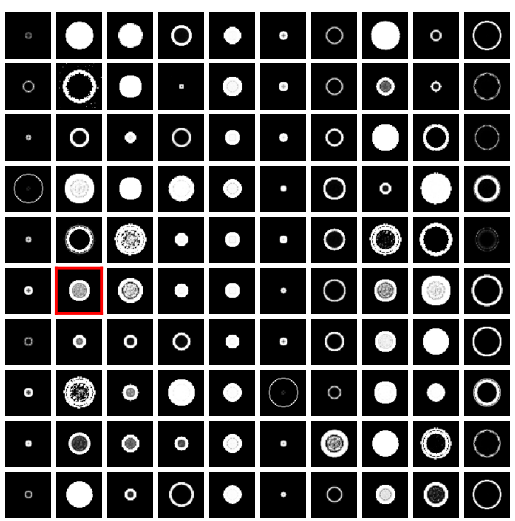
(a) VAE - $\mathcal{N}(0, I)$



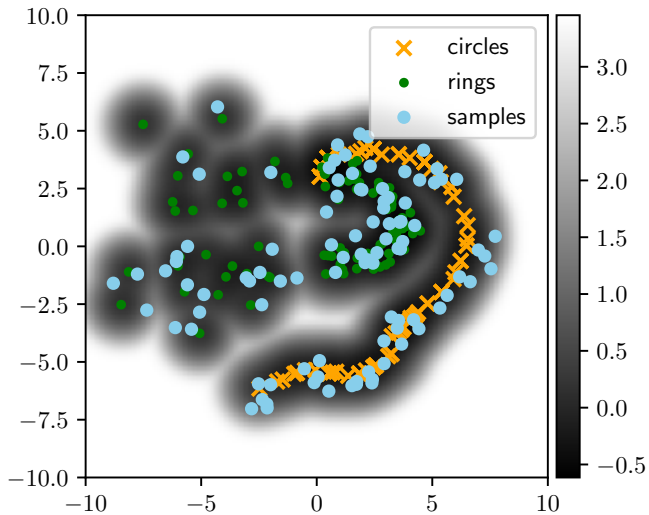


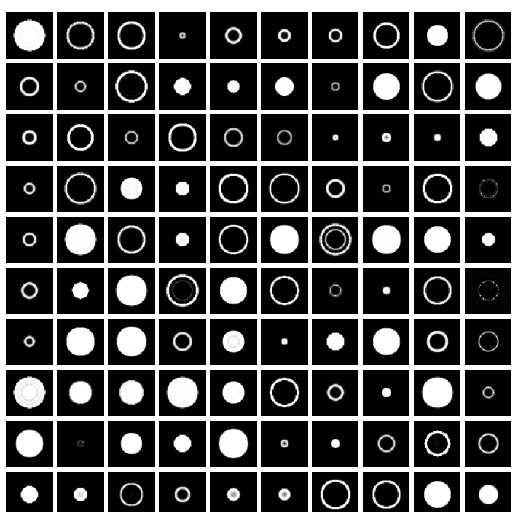
(b) VAE - VAMP (multimodal conditional prior)





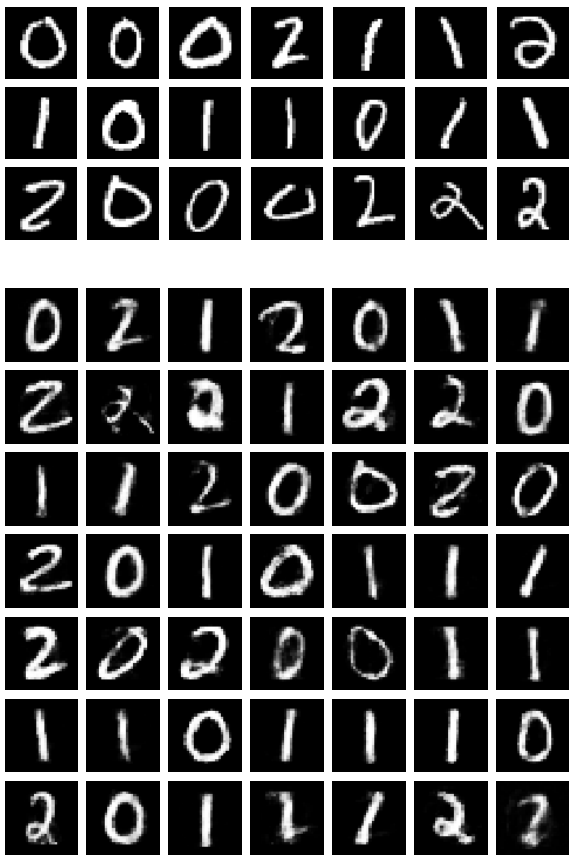
(c) Ours



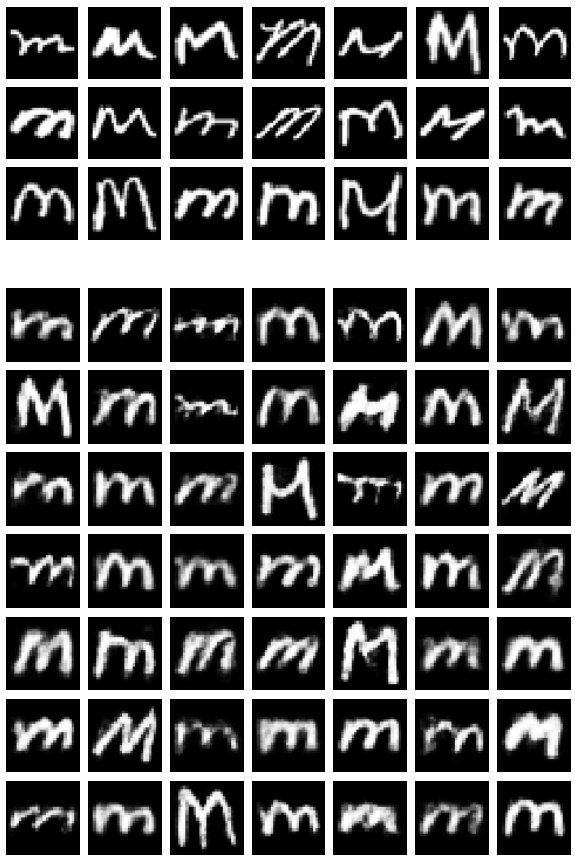


Sampling Comparison - Higher Dimension

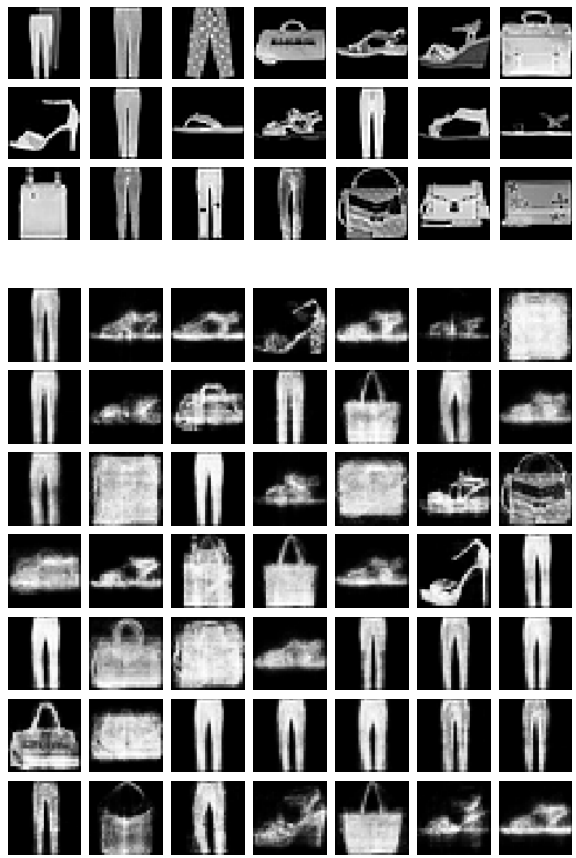
(a) *reduced* MNIST (120)



(b) *reduced* EMNIST (120)



(c) *reduced* Fashion (120)



Data Augmentation

1. Framework
2. Toy Data
3. Medical Imaging

Data Augmentation - Framework

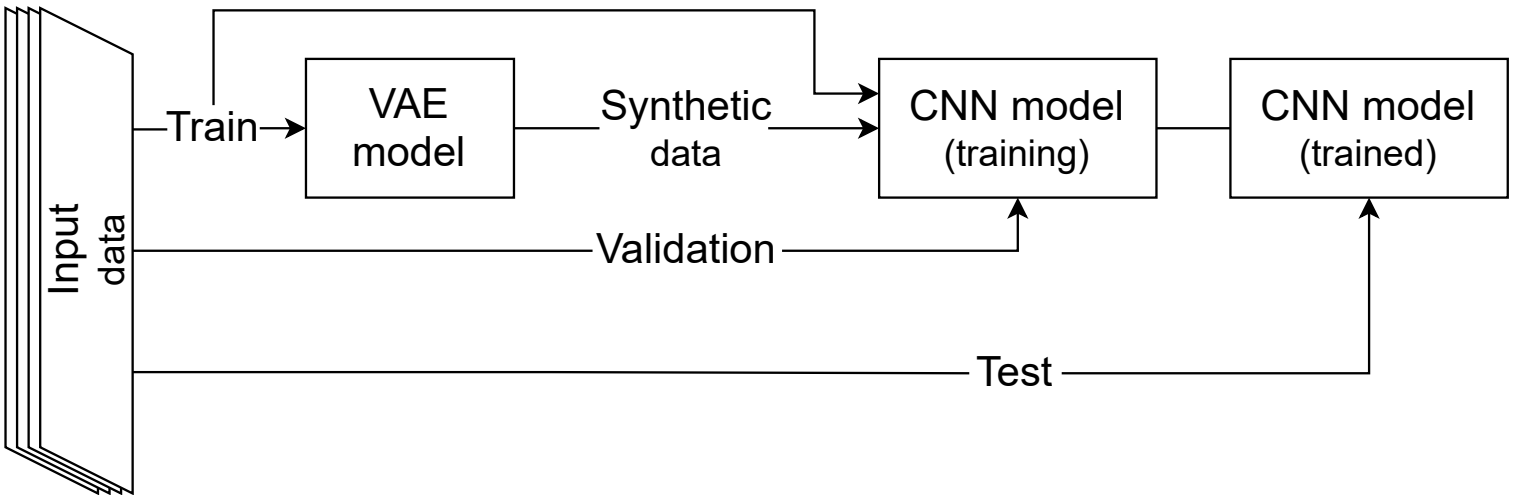


Figure: Data Augmentation pipeline

Performances are estimated using cross-validation.

Data Augmentation

1. Framework
2. Toy Data
3. Medical Imaging

Robustness Across Data Sets

Table: Classification results on *reduced* data sets ($\sim$ 50 samples per class)

	MNIST	MNIST (unbal.)	EMNIST (unbal.)	FASHION
Baseline	89.9 ± 0.6	81.5 ± 0.7	82.6 ± 1.4	76.0 ± 1.5
Baseline + Synthetic				
Basic Augmentation (X5)	92.8 ± 0.4	86.5 ± 0.9	85.6 ± 1.3	77.5 ± 2.0
Basic Augmentation (X10)	88.2 ± 2.2	82.0 ± 2.4	85.7 ± 0.3	79.2 ± 0.6
Basic Augmentation (X15)	92.8 ± 0.7	85.8 ± 3.4	86.6 ± 0.8	80.0 ± 0.5
VAE - 200*	88.5 ± 0.9	84.0 ± 2.0	81.7 ± 3.0	78.6 ± 0.4
VAE - 2k*	92.2 ± 1.6	88.0 ± 2.2	86.0 ± 0.2	79.3 ± 1.1
Ours-200	91.0 ± 1.0	84.1 ± 2.0	85.1 ± 1.1	77.0 ± 0.8
Ours-500	92.3 ± 1.1	87.7 ± 0.9	85.1 ± 1.1	78.5 ± 0.9
Ours-1k	93.2 ± 0.8	$\mathbf{89.7 \pm 0.8}$	87.0 ± 1.0	$\mathbf{80.2 \pm 0.8}$
Ours-2k	$\mathbf{94.3 \pm 0.8}$	89.1 ± 1.9	$\mathbf{87.6 \pm 0.8}$	78.1 ± 1.8

* Using a standard normal prior to generate

- Classic DA is data set dependent
- Vanilla VAE performs as well as classic DA

Robustness Across Data Sets

Table: Classification results on *reduced* data sets ($\sim$ 50 samples per class) **on synthetic samples only**

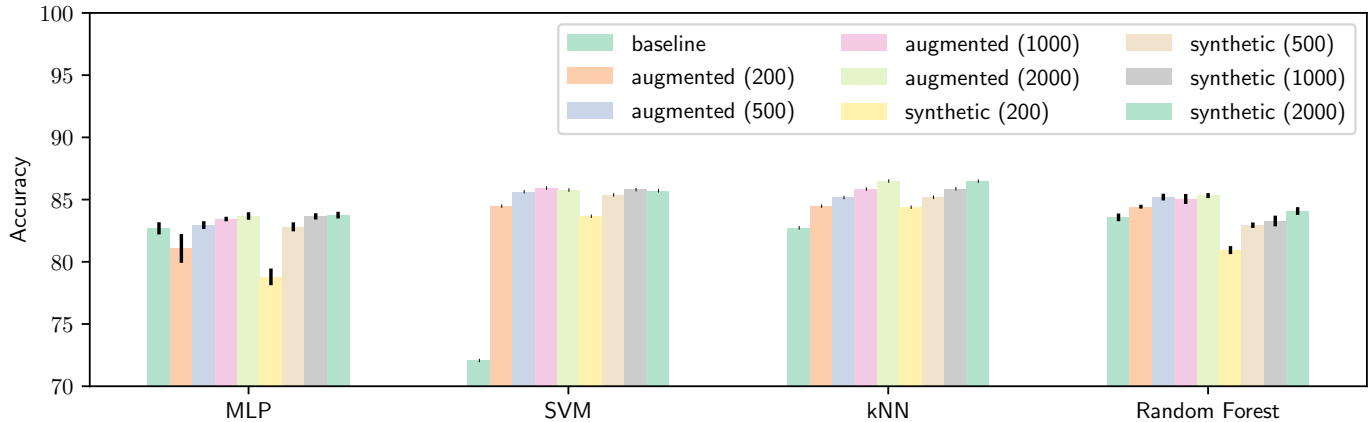
	MNIST	MNIST (unbal.)	EMNIST (unbal.)	FASHION
Baseline	89.9 ± 0.6	81.5 ± 0.7	82.6 ± 1.4	76.0 ± 1.5
Synthetic Only				
VAE - 200*	69.9 ± 1.5	64.6 ± 1.8	65.7 ± 2.6	73.9 ± 3.0
VAE - 2k*	86.5 ± 2.2	79.6 ± 3.8	78.8 ± 3.0	76.7 ± 1.6
Ours-200	87.2 ± 1.1	79.5 ± 1.6	77.0 ± 1.6	77.0 ± 0.8
Ours-500	89.1 ± 1.3	80.4 ± 2.1	80.2 ± 2.0	78.5 ± 0.8
Ours-1k	90.1 ± 1.4	86.2 ± 1.8	82.6 ± 1.3	79.3 ± 0.6
Ours-2k	92.6 ± 1.1	87.5 ± 1.3	86.0 ± 1.0	78.3 ± 0.9

* Using a standard normal prior to generate

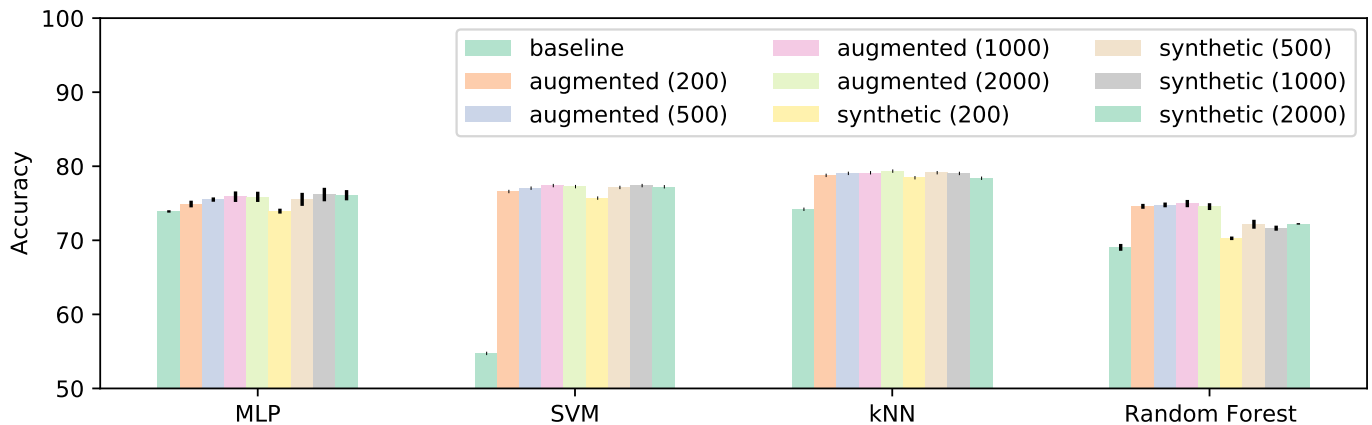
- The proposed model seems to create diverse samples relevant to the classifier

Robustness Across Classifiers

(a) *reduced* MNIST balanced



(b) *reduced* MNIST unbalanced



A Note on the Method Scalability

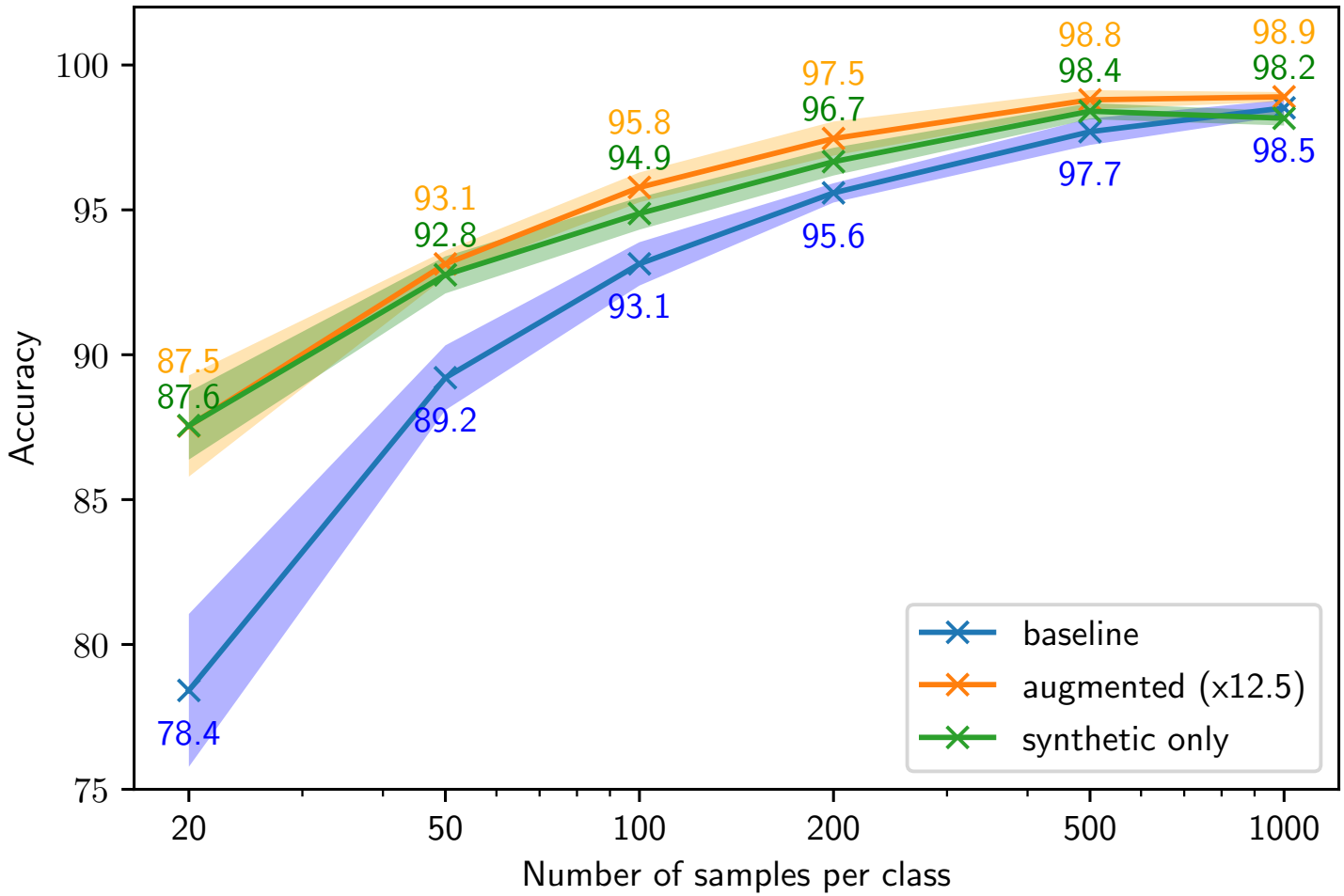


Figure: Benchmark classifier accuracy according to the number of samples in the training set on MNIST.

Data Augmentation

1. Framework
2. Toy Data
3. Medical Imaging

Datasets and classification task

Classification task: Alzheimer's disease patients (**AD**) vs Cognitively Normal participants (**CN**) using T1-weighted MR images.





Table: Summary of participant demographics, mini-mental state examination (MMSE) and global clinical dementia rating (CDR) scores at baseline.

Data set	Label	Obs.	Age	Sex M/F	MMSE	CDR
ADNI	CN	403	73.3 ± 6.0	185/218	29.1 ± 1.1	0: 403
	AD	362	74.9 ± 7.9	202/160	23.1 ± 2.1	0.5: 169, 1: 192, 2: 1
AIBL	CN	429	73.0 ± 6.2	183/246	28.8 ± 1.2	0: 406, 0.5: 22, 1: 1
	AD	76	74.4 ± 8.0	33/43	20.6 ± 5.5	0.5: 31, 1: 36, 2: 7, 3: 2

MRI preprocessing

Bias field correction (N4ITK) + linear registration (ANTS) + cropping

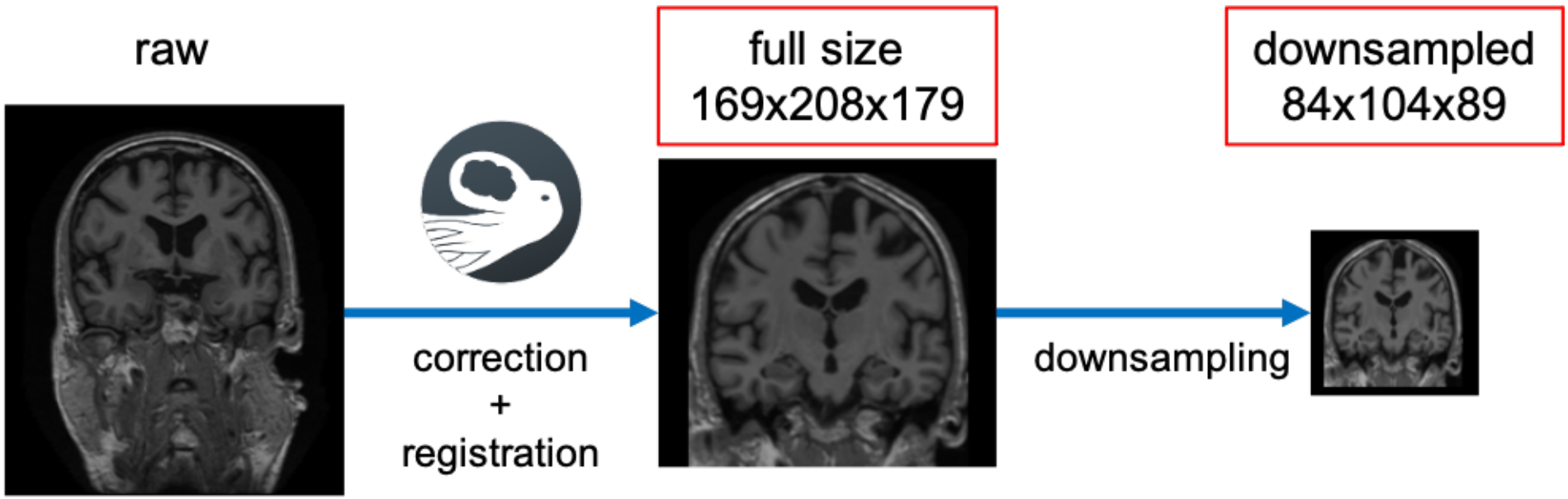


Figure: Preprocessed MRI used in the study

Find wonderful data at:
`/network/lustre/dtlake01/aramis/datasets/adni/caps/caps_v2021`

Synthesized images

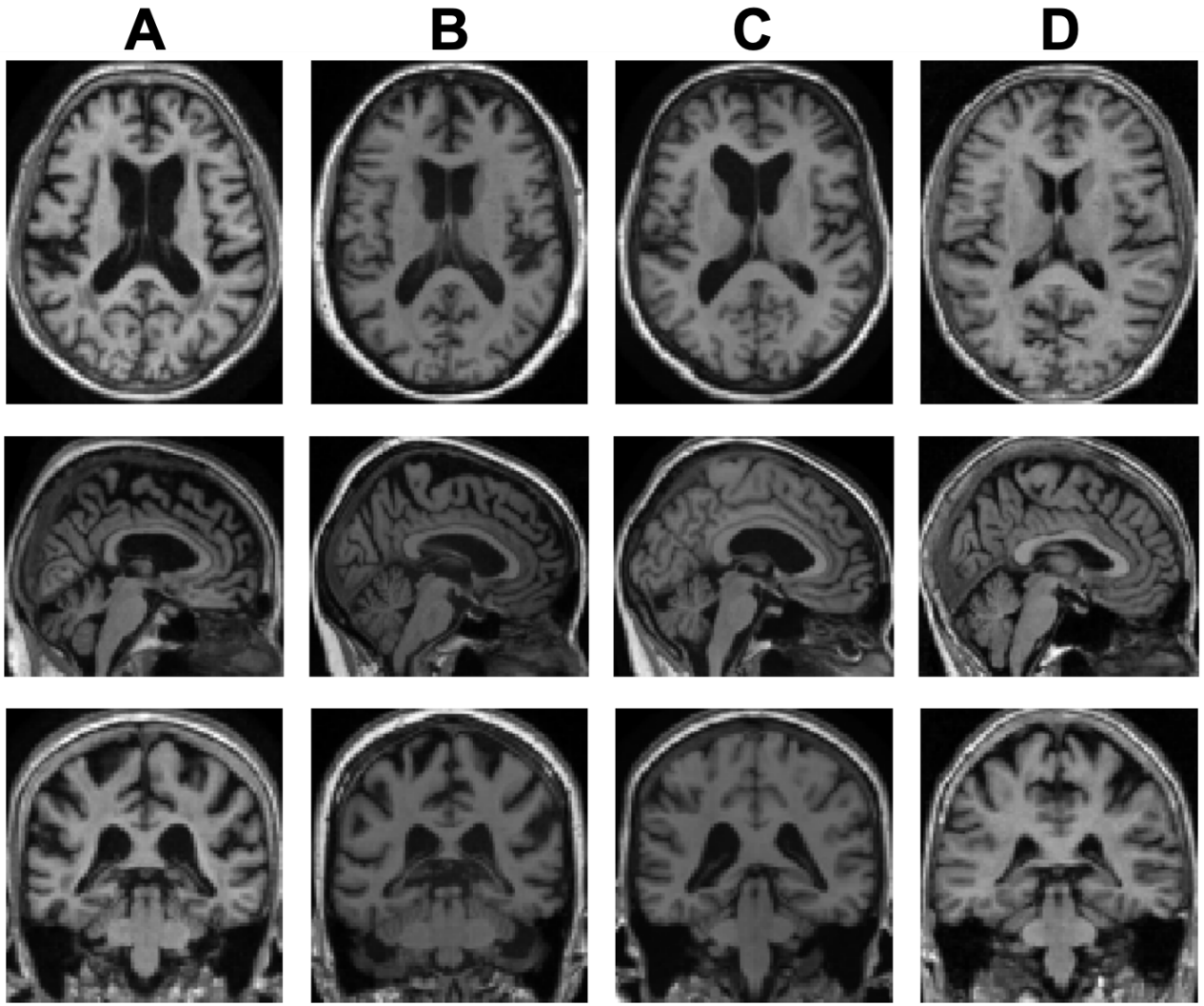


Figure: Example of two *true* patients compared to two generated by our method. Can you find the intruders ?

Synthesized images

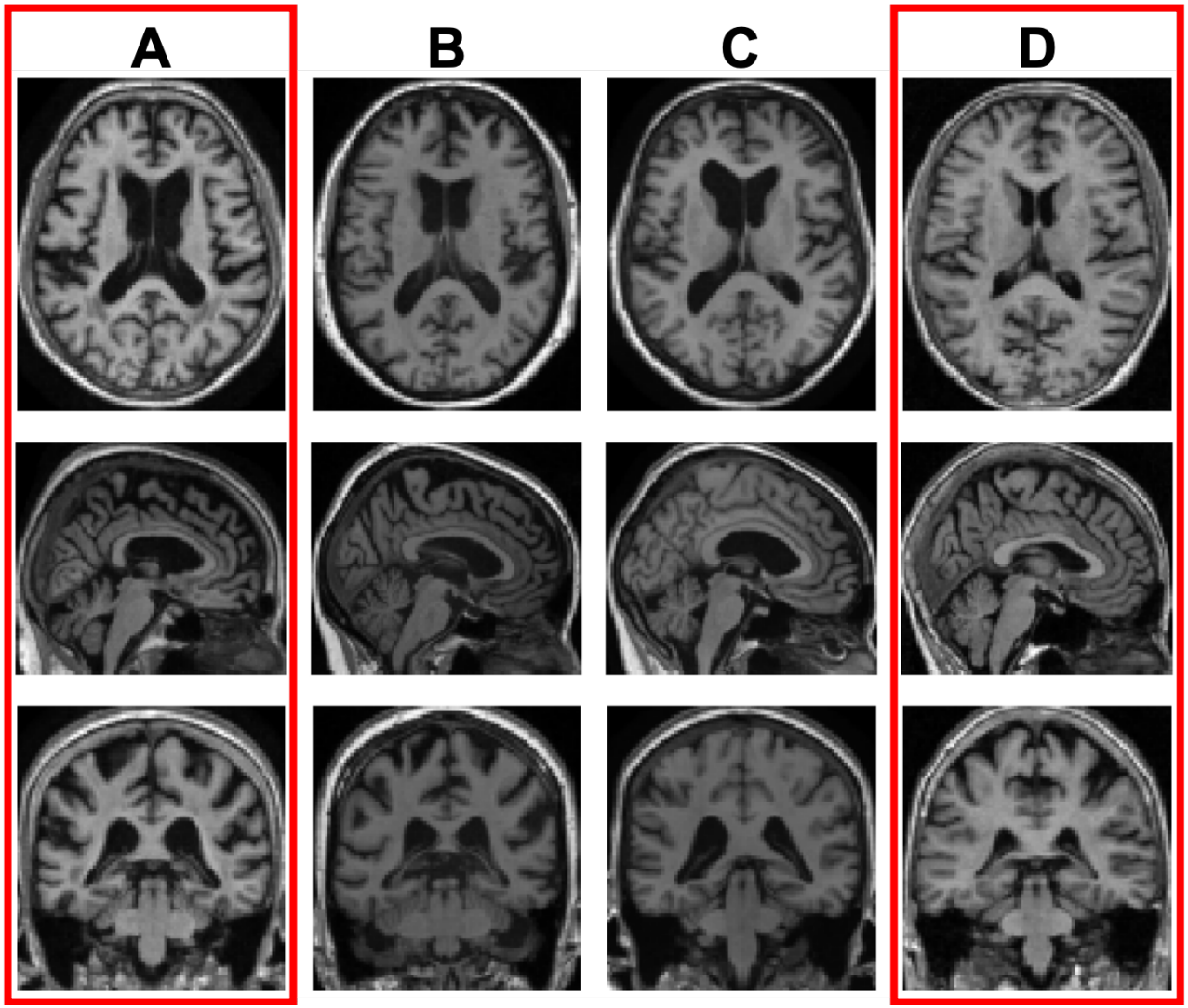
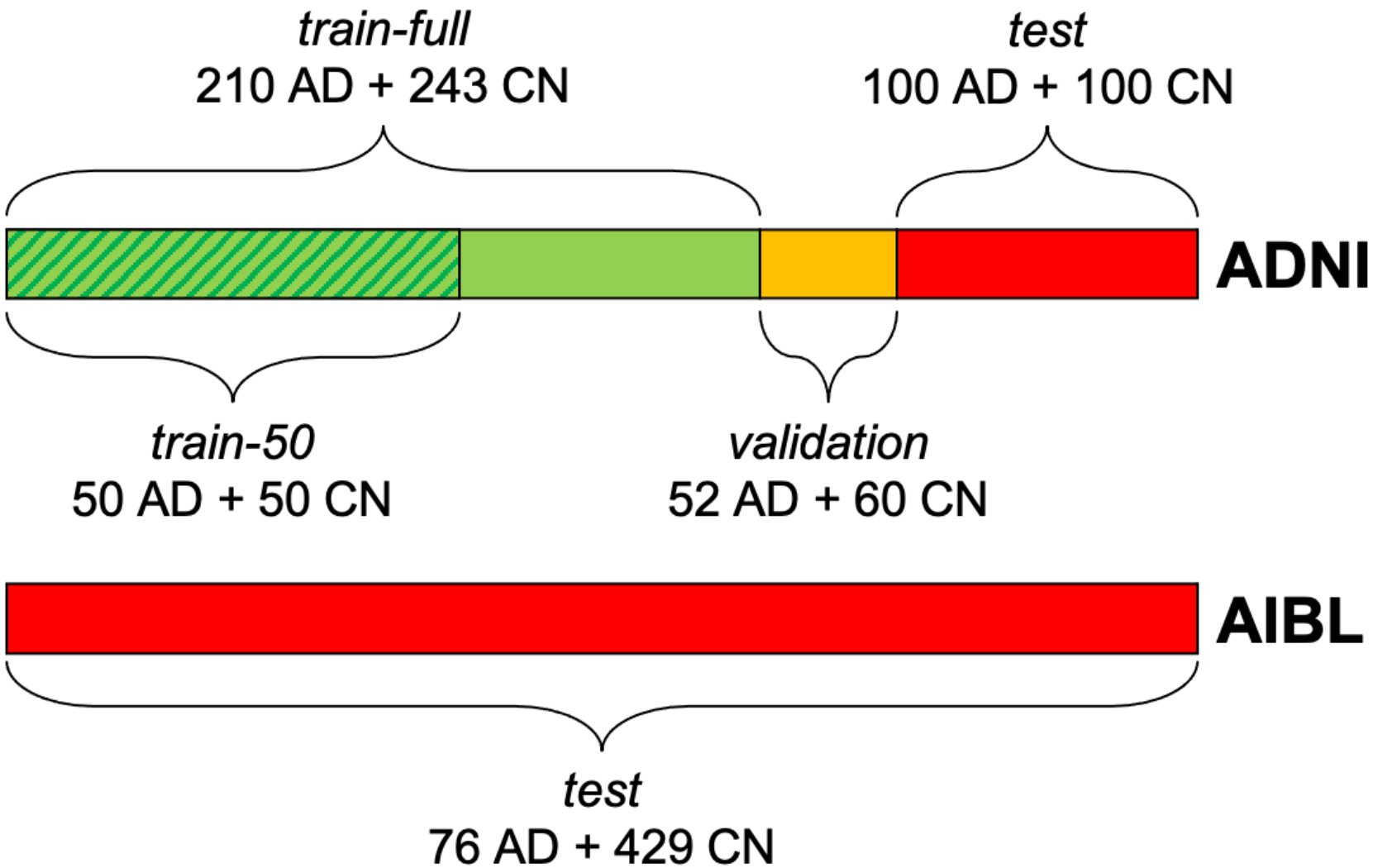


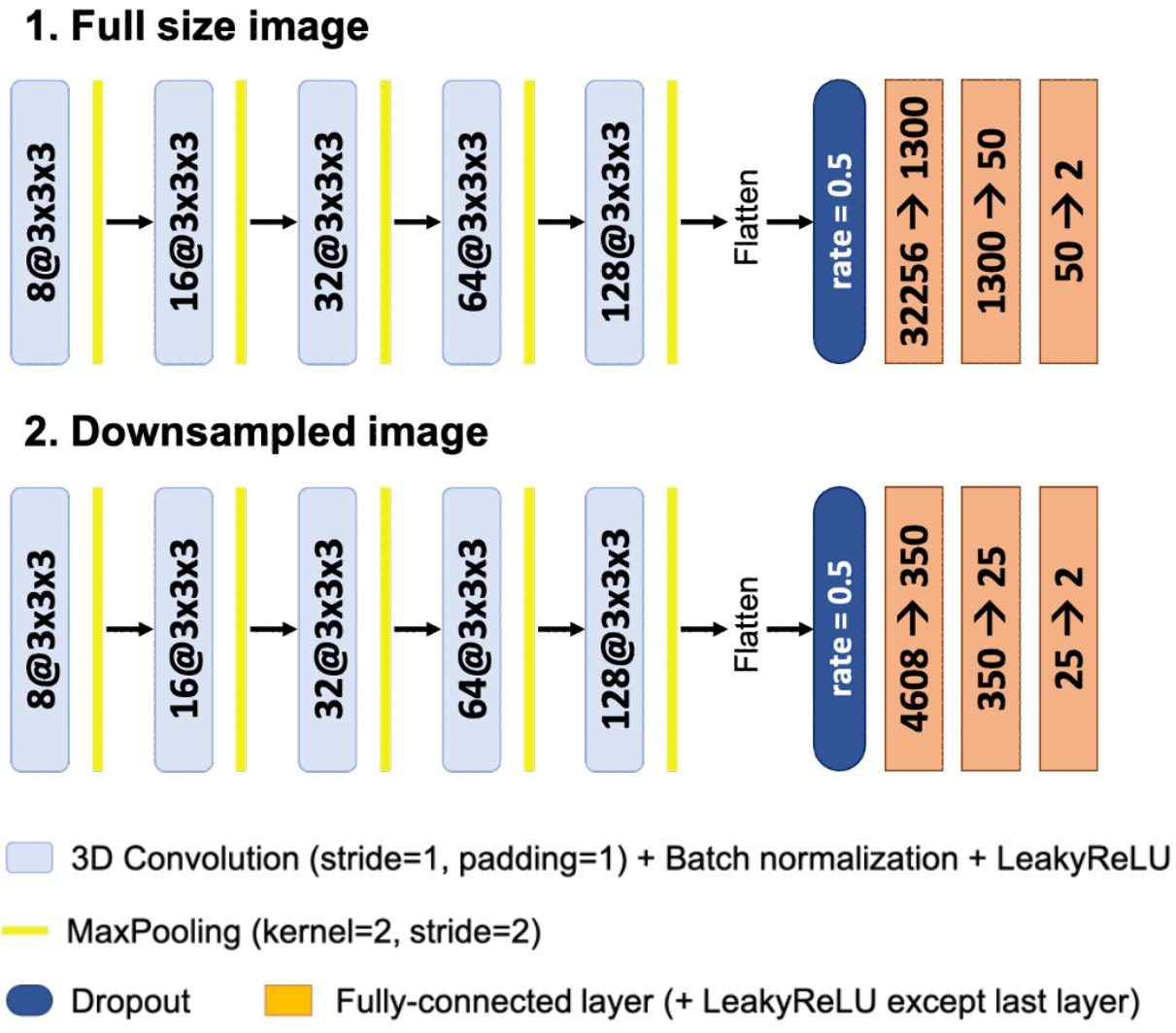
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Evaluation procedure



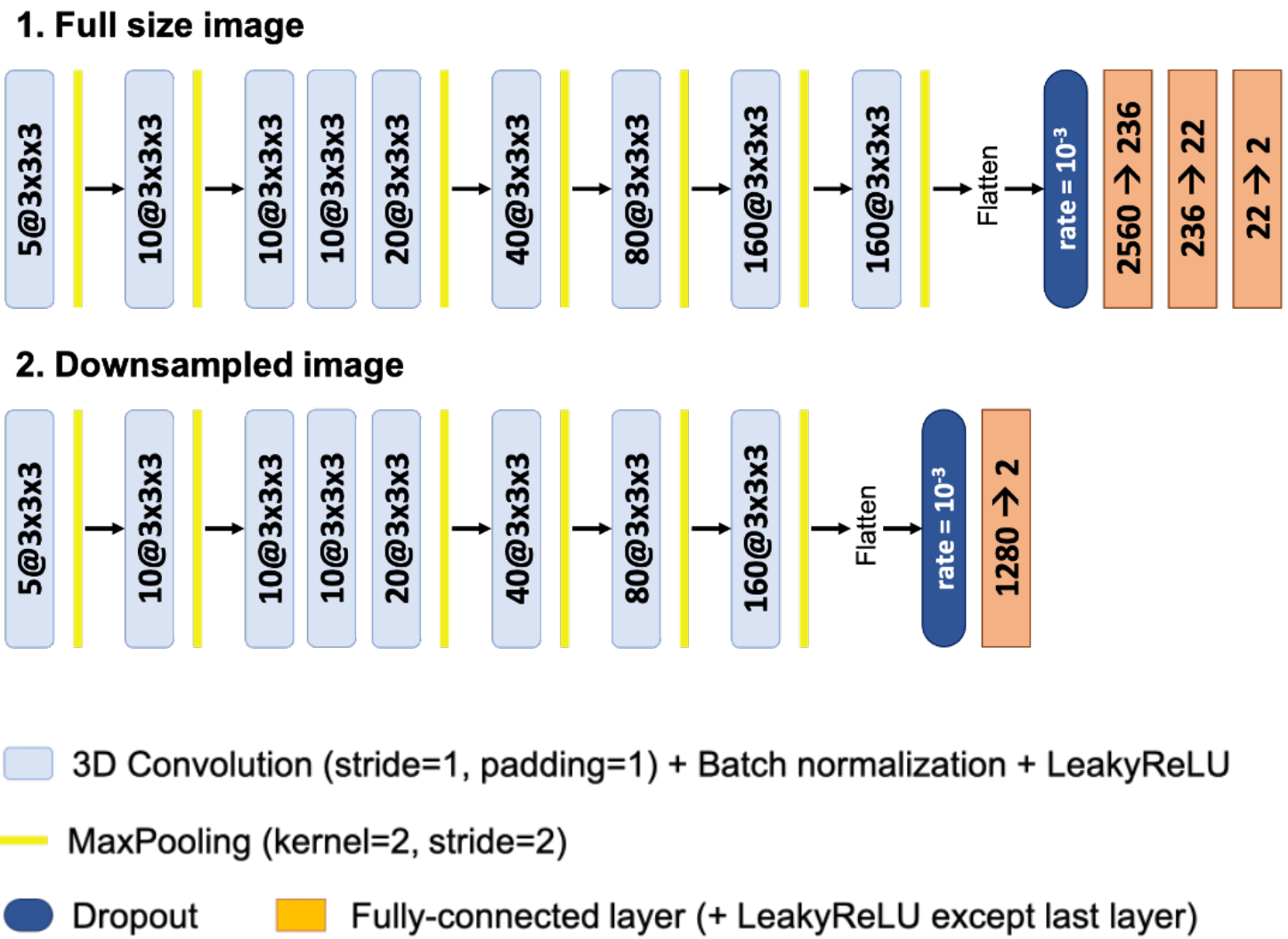
CNN architectures for classifier

Baseline architectures provided by a previous study [WTSDM+20]



CNN architectures for classifier

Optimized architectures opitmize with random search procedure for this training set (ClinicaDL)



Experiments

Four series of experiments:

- **baseline** architecture on *train-50*
- **baseline** architecture on *train-full*
- **optimized** architecture on *train-50*
- **optimized** architecture on *train-full*

For each experiment 20 CNNs are run and the performance is the mean value of the 20 performance values.

Results on train-50 with baseline CNN

Table: Mean test performance of each series of 20 runs trained with the **baseline** hyperparameters on *train-50* set.

data set	ADNI balanced accuracy	AIBL balanced accuracy
real	66.3 ± 2.4	67.2 ± 4.1
real (high-resolution)	67.9 ± 2.3	66.5 ± 3.0
500 synthetic + real	69.4 ± 1.6	68.5 ± 2.5
1000 synthetic + real	70.5 ± 2.1	70.6 ± 3.1
2000 synthetic + real	71.2 ± 1.6	72.8 ± 2.2
3000 synthetic + real	72.6 ± 1.6	73.6 ± 3.0
5000 synthetic + real	$\mathbf{74.1 \pm 2.2}$	$\mathbf{76.1 \pm 3.6}$
10000 synthetic + real	74.0 ± 2.7	74.9 ± 3.2

Increase of balanced accuracy of 6.2 points on ADNI and 8.9 points on AIBL

Results on train-full with baseline CNN

Table: Mean test performance of each series of 20 runs trained with the **baseline** hyperparameters on *train-full* set.

data set	ADNI balanced accuracy	AIBL balanced accuracy
real	77.7 ± 2.5	78.4 ± 2.4
real (high-resolution)	80.6 ± 1.1	80.4 ± 2.6
500 synthetic + real	82.2 ± 2.4	82.9 ± 2.5
1000 synthetic + real	84.4 ± 1.8	83.7 ± 2.3
2000 synthetic + real	85.9 ± 1.6	83.8 ± 2.2
3000 synthetic + real	85.8 ± 1.7	84.4 ± 1.8
5000 synthetic + real	85.7 ± 2.1	84.2 ± 2.2
10000 synthetic + real	$\mathbf{86.3 \pm 1.8}$	$\mathbf{85.1 \pm 1.9}$

Increase of balanced accuracy of 5.7 points on ADNI and 4.7 on AIBL

Results on train-50 with optimized CNN

Table: Mean test performance of each series of 20 runs trained with the **optimized** hyperparameters on *train-50* set.

data set	ADNI balanced accuracy	AIBL balanced accuracy
real	75.5 ± 2.7	75.6 ± 4.1
real (high-resolution)	72.1 ± 3.1	71.2 ± 5.1
500 synthetic + real	75.6 ± 2.5	76.0 ± 4.2
1000 synthetic + real	77.8 ± 2.3	80.9 ± 3.2
2000 synthetic + real	76.9 ± 2.4	80.0 ± 3.6
3000 synthetic + real	77.8 ± 1.9	81.2 ± 3.7
5000 synthetic + real	76.9 ± 2.5	80.9 ± 2.7
10000 synthetic + real	$\mathbf{78.0 \pm 2.1}$	$\mathbf{81.9 \pm 2.2}$

Increase of balanced accuracy of 2.5 points on ADNI and 6.3 points on AIBL

Results on train-full with optimized CNN

Table: Mean test performance of each series of 20 runs trained with the **optimized** hyperparameters on *train-full* set.

data set	ADNI balanced accuracy	AIBL balanced accuracy
real	85.5 ± 2.4	81.9 ± 3.2
real (high-resolution)	85.7 ± 2.5	84.4 ± 1.7
500 synthetic + real	86.0 ± 1.8	83.2 ± 2.4
1000 synthetic + real	86.5 ± 1.9	83.7 ± 2.0
2000 synthetic + real	$\mathbf{87.2 \pm 1.7}$	84.0 ± 2.0
3000 synthetic + real	85.8 ± 2.6	83.6 ± 3.2
5000 synthetic + real	86.4 ± 1.3	83.5 ± 2.2
10000 synthetic + real	86.7 ± 1.8	$\mathbf{84.3 \pm 1.8}$

Increase of balanced accuracy of 1.5 point on ADNI and -0.1 point on AIBL

Conclusion

We have proposed

- a new geometry aware VAE-based data augmentation framework relevant for representing and classifying data in the HDLSS setting.
- Validated on classification tasks on *toy* and *real-life* data sets in particular in the *High dimension low sample size setting*.

Conclusion

We have proposed

- a new geometry aware VAE-based data augmentation framework relevant for representing and classifying data in the HDLSS setting.
- Validated on classification tasks on *toy* and *real-life* data sets in particular in the *High dimension low sample size setting*.

Strengths:

- Independent on the nature of the data set: from 2D images (MNIST, EMNIST, FASHION) to 3D medical images (ADNI and AIBL),
- Relevant synthetic data: classifiers achieved a similar or better classification performance when trained only on synthetic data than on the *real* train set.
- Classifier independence: MLP, random forest, k-NN and SVM (on toy data sets) ; baseline and optimized parameters (on medical images).

Conclusion

We have proposed

- a new geometry aware VAE-based data augmentation framework relevant for representing and classifying data in the HDLSS setting.
- Validated on classification tasks on *toy* and *real-life* data sets in particular in the *High dimension low sample size setting*.

Limitations - what could be improved:

- No extensive search on VAE architecture.
- Would it benefit from the use of longitudinal data?
- *train-50* is still large compared to some medical data sets...

Implementation available
https://clementchadebec.github.io/projects/

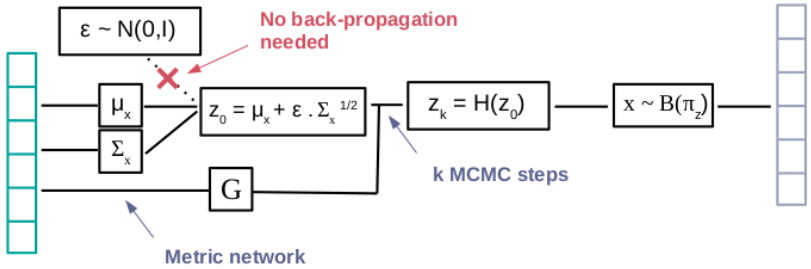
PyRaug

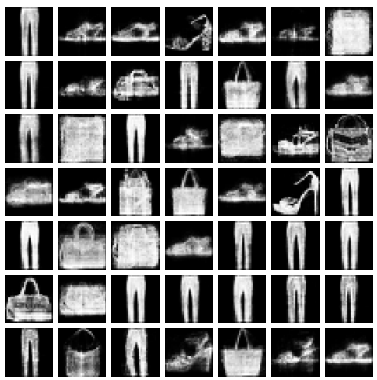
AND Extensive comparison of data generation based on VAEs

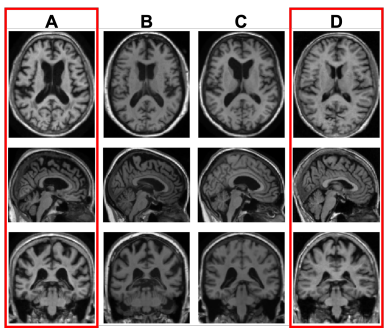
Models	Training example	Paper	Official Implementation
Autoencoder (AE)	Open in Colab		
Variational Autoencoder (VAE)	Open in Colab	link	
Beta Variational Autoencoder (BetaVAE)	Open in Colab	link	
Disentangled Beta Variational Autoencoder (DisentangledBetaVAE)	Open in Colab	link	
Disentangling by Factorising (FactorVAE)	Open in Colab	link	
Beta-TC-VAE (BetaTCVAE)	Open in Colab	link	link
Importance Weighted Autoencoder (IWAE)	Open in Colab	link	link
VAE with perceptual metric similarity (MSSSIM_VAE)	Open in Colab	link	
Wasserstein Autoencoder (WAE)	Open in Colab	link	link
Info Variational Autoencoder (INFOVAE_MMD)	Open in Colab	link	
VAMP Autoencoder (VAMP)	Open in Colab	link	link
Hyperspherical VAE (SVAE)	Open in Colab	link	link
Adversarial Autoencoder (Adversarial_AE)	Open in Colab	link	
Variational Autoencoder GAN (VAEGAN)	Open in Colab	link	link
Vector Quantized VAE (VQVAE)	Open in Colab	link	link
Hamiltonian VAE (HVAE)	Open in Colab	link	link
Regularized AE with L2 decoder param (RAE_L2)	Open in Colab	link	link
Regularized AE with gradient penalty (RAE_GP)	Open in Colab	link	link
Riemannian Hamiltonian VAE (RHVAE)	Open in Colab	link	link



Thank you!







https://clementchadebec.github.io/projects/

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- Maxime Louis, *Computational and statistical methods for trajectory analysis in a Riemannian geometry setting*, PhD Thesis, Sorbonnes universités, 2019.
- Radford M Neal and others, *MCMC using hamiltonian dynamics*, Handbook of Markov Chain Monte Carlo **2** (2011), no. 11, 2.
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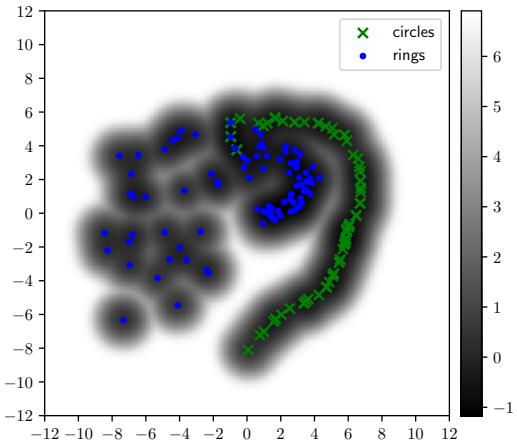
Danilo Jimenez Rezende, Shakir Mohamed, and Daan Wierstra, *Stochastic backpropagation and approximate inference in deep generative models*, International conference on machine learning, PMLR, 2014, pp. 1278–1286.



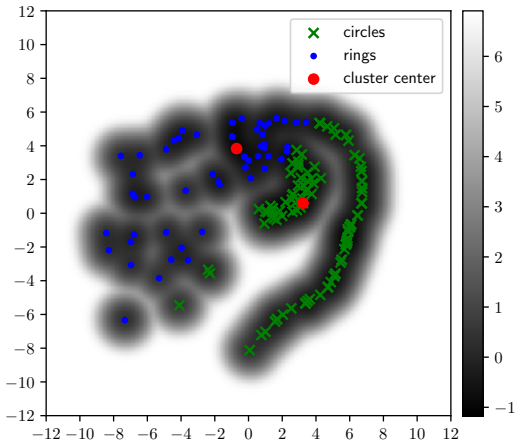
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Clustering

True labels



Euclidean k-medoids



Riemannian k-medoids

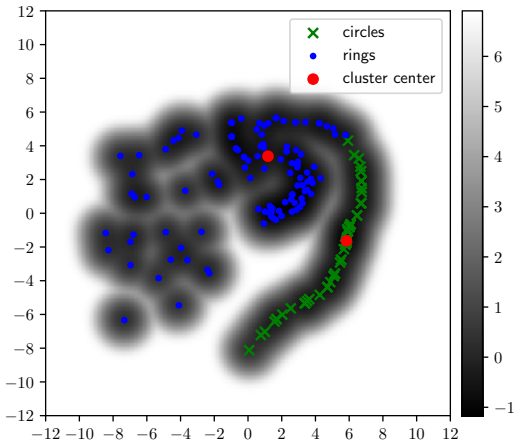
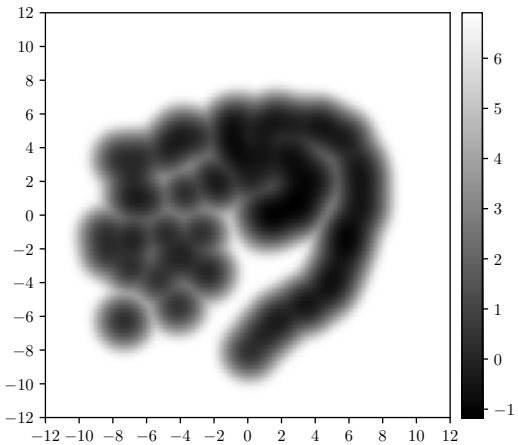
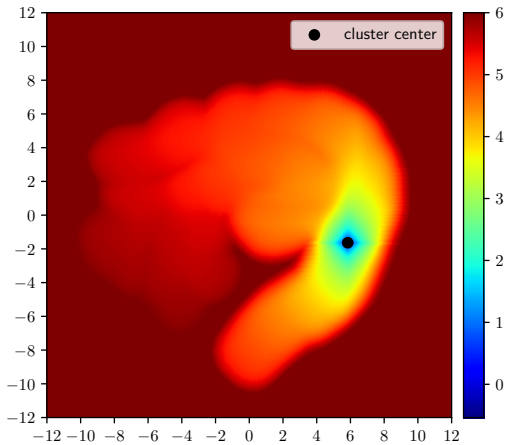


Figure: Euclidean and Riemannian k-medoids custering.





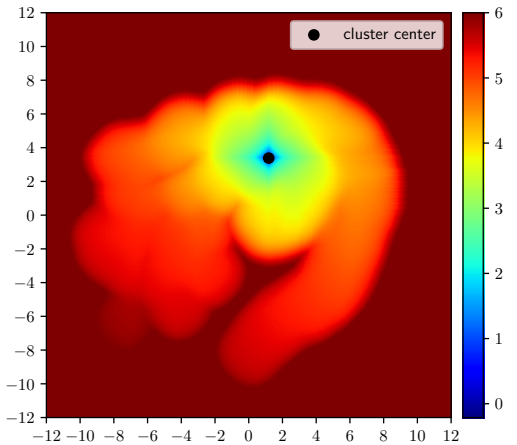


Figure: Distance maps.

Results - Clustering

Data set	Model	Subset 1	Subset 2	Subset 3	Mean
Synthetic data	linear	53.88	62.52	71.63	62.68
	geodesic	**71.41**	**81.39**	**79.49**	**77.43**
MNIST 1	linear	89.73	93.11	91.80	91.55
	geodesic	**91.68**	**94.51**	**95.63**	**93.94**
MNIST 2	linear	68.24	69.22	79.05	71.17
	geodesic	**70.35**	**71.34**	**79.64**	**73.78**
MNIST 3	linear	75.55	75.76	81.70	77.67
	geodesic	**76.08**	**77.94**	**81.96**	**78.66**
FashionMNIST 1	linear	90.47	91.63	86.78	89.63
	geodesic	**91.44**	**92.55**	**87.46**	**90.48**
FashionMNIST 2	linear	92.20	91.26	93.30	92.25
	geodesic	**93.56**	**91.80**	**94.12**	**93.16**
FashionMNIST 3	linear	72.46	79.58	83.16	78.40
	geodesic	**74.89**	**81.88**	**84.83**	**80.53**

Table: F1-Scores.